L1.9 L'Hôpital's Rule

Work in your group. Probe every limit before you touch it.

Where L'Hôpital's Rule comes from

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1. Two functions are both zero at $x = 3$. Their slopes there are $f'(3) = 10$ and $g'(3) = -4$.

(a) Step off $x = 3$ by a nudge dx. Write df and dg.

(b) What is $\lim_{x \to 3} \dfrac{f(x)}{g(x)}$?

(c) Your answer to (b) has no dx in it. Why does that matter?

2. Evaluate each of these. Every one of them probes to $\dfrac{0}{0}$.

(a) $\lim_{x \to 0} \dfrac{5x}{x}$

(b) $\lim_{x \to 0} \dfrac{x^2}{x}$

(c) $\lim_{x \to 0} \dfrac{x}{x^3}$

Three identical forms, three different answers. Write one sentence saying what $\dfrac{0}{0}$ tells you — and what it does not.

The two direct forms: $\frac{0}{0}$ and $\frac{\infty}{\infty}$

3. $\lim_{x\to 0} \frac{\sin 3x}{x}$

4. $\lim_{x\to\infty} \frac{2x^2-5x+1}{6x^2+x+3}$

5. $\lim_{x\to -2} \frac{x+2}{x^2+3x+2}$ — do it **two ways**, and say which you would use if you met it cold.

6. $\lim_{x\to 0} \frac{x^2}{1-\cos x}$

> **Stop.** Check the form again after every application. When it stops being indeterminate, stop.

7. $\lim_{x\to\infty} \frac{\sqrt{4x^2+1}}{3x-1}$

One application does not finish this one — it hands you a limit at infinity that you still have to evaluate.

8. For which of these is L'Hôpital's Rule **allowed**? Probe each one first, and write the form you got.

(a) $\lim_{x \to 0} \dfrac{1 - \cos x}{x^2}$

(b) $\lim_{x \to 0} \dfrac{\cos x}{x^2}$

(c) $\lim_{x \to \pi} \dfrac{\sin x}{x - \pi}$

(d) $\lim_{x \to 1} \dfrac{x^2 + 3}{x + 1}$

At the boards

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9. BOARDS Build a limit to order.

(a) Write a limit as $x \to 0$ that probes to $\dfrac{0}{0}$ and whose value is **7**.

(b) Now one that probes to $\dfrac{0}{0}$ and whose value is **0**.

(c) Now one that probes to $\dfrac{0}{0}$ and needs the rule **twice**.

Check each other's. A different group's answer should not look like yours.